COURSE DESCRIPTION			
Program	**Automobile Engineering**		
Course Title	**Fundamentals of Internal Combustion Engines**		
Course Code	**2052**	**Semester**	**II**
Type of Course	**Theory**	**Contact Hours**	**60**
Teaching Scheme (L: T:P:R)	**4:0:0:0**	**Credits**	**4**
CIE Marks	**40**	**ESE Marks**	**60**

Introduction

Fundamentals of IC Engines is dedicated to the study of the construction, operational principles, and systemic interaction of internal combustion engine components utilized in automotive applications. The curriculum encompasses valve mechanisms, engine lubrication and cooling systems, and related engine subsystems, thereby establishing a robust foundation for subsequent advanced coursework in automobile engineering. The course also introduces engineering practices related to component selection, maintenance considerations, environmental aspects and effective technical communication in engine systems.

Course Objectives

1	To understand the construction, materials and manufacturing aspects of basic engine components such as cylinder block, head, piston and liners.
2	To understand the construction, materials and working of reciprocating and rotating engine components including crankshaft, camshaft, bearings and flywheel.
3	To understand the construction and operation of valve mechanisms, cylinder head types, cam arrangements and modern valve timing systems.
4	To understand the principles, components and working of engine lubrication and cooling systems.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Classification of IC engine, Operation of two-stroke and four-stroke internal combustion engines	1051	Basic Automobile Engineering	I

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Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Explain construction, materials and manufacturing methods of engine components.	Understand
CO2	Explain working principles and interaction of reciprocating and rotating engine components.	Understand
CO3	Explain valve mechanisms, cylinder head types and modern valve timing systems used in IC engines.	Understand
CO4	Explain and compare lubrication and cooling systems considering efficiency, reliability, maintenance and environmental aspects of engine operation.	Understand

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	-	-	-	-	-	-	-	2
CO2	3	2	2	-	-	2	-	-	-	-	2
CO3	3	2	2	2	2	2	2	-	2	2	2
CO4	3	2	-	2	-	2	2	-	-	-	2
Total	12	8	4	4	2	6	4	-	2	2	8
Correlation Level	3	2	2	2	2	2	2	-	2	2	2
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"											

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
4	0	0	0	6	60	4	40	60	100	0	0	0	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus – Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Engine Construction Details - I	Introduction to engine construction; cylinder block and crankcase, cylinder head, oil pan, crank case ventilation and its role in emission control, intake and exhaust manifolds, gaskets, cylinder liners (dry and wet), piston (construction, materials and types), piston rings.	15	0
Module II	Engine Construction Details - II	Connecting rod, piston pin, Crankshaft (construction and materials), Engine Bearings, Camshaft (construction and materials), Camshaft drives, Vibration dampers, Flywheel, Types of Flywheel.	15	0
Module III	Types of cylinder head and valve mechanism	Types of Cylinder Heads (L, T, F, I Heads), Poppet valve (construction and materials), Sodium cooled valves, valve train components, Valve actuating mechanisms, Camshaft arrangements (SOHC & DOHC), Valve timing diagram of four-stroke engine (theoretical and actual), Variable Valve Timing (VVT) – Concept, Need and Types of VVT.	15	0
Module IV	Automobile Engine Cooling & Lubrication	Concept and methods of lubrication, properties of lubricating oil, single-grade and multi-grade oils, Oil additives, Engine lubrication systems, Types of oil pumps, Oil filtering systems, Oil cooler and pressure relief valve, Engine Cooling systems (air and water cooling), Cooling system components, Radiator cap, Expansion reservoir, and antifreeze solutions.	15	0

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Introduction to Engine Construction	Discuss the basics of automotive engines and list engine components	L	CO1	Understand	1
Cylinder Block & Crankcase	Discuss the constructional details, materials and manufacturing methods	L	CO1	Remember	2
Cylinder Head	Explain combustion chamber types and materials used	L	CO1	Understand	2
Oil Pan & Crankcase Ventilation	Describe the design and function of the oil pan. Detail various crankcase ventilation systems and analyze their impact on pollution.	L	CO1	Understand	2
Intake & Exhaust Manifolds	Explain functions and manifold design	L	CO1	Understand	1
Gaskets	Detail different types of gaskets and their materials.	L	CO1	Understand	1
Cylinder Liners	Explain dry and wet liners, and compare dry and wet liners	L	CO1	Understand	2
Piston and its types	Explain the functions, constructional features, materials, and special types of pistons such as T-slot piston, cam-ground piston, taper piston, wire-wound piston, offset piston, and autothermic piston used in internal combustion engines.	L	CO1	Understand	2
Piston Rings	Discuss the functions, materials, and types of piston rings, including compression rings and oil control rings, and classify piston rings based on piston ring end gap.	L	CO1	Understand	2
Module II					
Connecting Rod	Explain the construction, materials, manufacturing and functions of connecting rod.	L	CO2	Understand	2

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Piston Pin	Describe the piston pin: materials used and arrangement methods (set screw type, semi-floating, and fully floating).	L	CO2	Understand	2
Crankshaft – Construction And materials	Discuss the constructional details and functions of the crankshaft, manufacturing method, and materials used	L	CO2	Understand	2
Engine Bearings	Detail types (standard duty, medium duty, Heavy duty, Extra heavy duty) requirements and materials functions of engine bearings.	L	CO2	Understand	2
Camshaft – Construction and Materials	Discuss the Functions, constructional details, and materials used for the camshaft.	L	CO2	Understand	1
Camshaft Drives	Detail different types of camshaft drives, such as gear, chain, and belt drives.	L	CO2	Understand	2
Vibration Dampers	Explain the need and working of vibration dampers.	L	CO2	Understand	1
Flywheel	Discuss the function, construction, and need of a flywheel	L	CO2	Understand	1
Types of Flywheel	Classify and explain solid disc, rim type, and dual mass flywheel	L	CO2	Understand	2
Module III					
Types of Cylinder Heads (L, T, F, I Heads)	Explain the construction and arrangement of valves in L-head, T-head, F-head, and I-head engines	L	CO3	Understand	2
Poppet Valve – Construction and Materials,	Explain the construction and materials of the poppet valve, valve seat, and valve guide	L	CO3	Understand	2
Sodium cooled valves	Explain sodium-cooled valves	L	CO3	Understand	1
Valve Train Components	Explain the construction and functions of valve train components such as valve tappet, tappet clearance, push rod, rocker arm, and valve rotators	L	CO3	Understand	2
Valve Actuating Mechanisms	Discuss OHV and OHC systems	L	CO3	Understand	2
Camshaft Arrangements	Explain SOHC and DOHC comparison	L	CO3	Understand	1
Valve Timing Diagram	Explain the theoretical and actual valve timing diagram of a 4-stroke engine	L	CO3	Understand	2
Variable Valve Timing (VVT) – Concept & Need	Explain the working principle and advantages of VVT	L	CO3	Understand	1
Types of VVT Systems	Explain Cam Phasing VVT, Continuous VVT (CVVT), and Variable Valve Lift & Timing (VTEC concept)	L	CO3	Understand	2
Module IV					
Various methods of engine lubrication.	Explain and justify methods of lubrication in engines.	L	CO4	Understand	1
Properties of Lubricating Oil	Explain the properties and requirements of lubricating oil used in internal combustion engines.	L	CO4	Understand	1
Single and Multi- Grade Oils and Oil Additives	Explain the concept and classification of single-grade and multi-grade oils and the purpose and types of oil additives.	L	CO4	Understand	1
Engine Lubrication Systems	Explain the working of the petroil, splash, pressure, dry sump, and wet sump lubrication systems.	L	CO4	Understand	2
Types of Oil Pumps	Explain the construction and working of gear pump, vane pump, and rotor pump.	L	CO4	Understand	2
Oil Filtering Systems	Explain the layout and working of full-flow and bypass flow oil filtering systems.	L	CO4	Understand	1
Oil Cooler, Oil Filter, and Pressure Relief Valve	Explain the functions of the oil cooler, pressure relief valve, and cartridge-type oil filter used in internal combustion engines.	L	CO4	Understand	2
Cooling System Types and Circulation Methods	Explain air-cooling and water-cooling systems, including thermosyphon and pump circulation methods.	L	CO4	Understand	2
Cooling System Components	Explain the construction and working of water pump, radiator, radiator core types (tubular and cellular), and thermostat (wax pellet and bellows types).	L	CO4	Understand	2

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Radiator Cap, Expansion Reservoir and Antifreeze solutions	Explain the functions of radiator pressure cap, expansion reservoir and the purpose of anti-freeze solutions.	L	CO4	Understand	1

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
	Module I	
1	Study manufacturing methods of cylinder block and cylinder head and analyze their effect on engine performance, cost and sustainability.	5
2	Prepare comparison chart of dry and wet cylinder liners considering performance, maintenance requirements and environmental impact.	4
3	Sketch and classify different types of pistons and explain their applications in automobile engines.	5
4	Case study on aluminium vs cast iron engine block construction considering strength, weight, manufacturing cost and sustainability.	4
5	Case study on aluminium vs cast iron engine block construction considering strength, weight, manufacturing cost and sustainability.	4
	Module II	
6	Draw crankshaft layout showing journals and crankpins and explain its importance for engine reliability and safe operation.	5
7	Study bearing clearance and oil film formation in engine bearings.	4
8	Compare types of flywheel with applications considering durability, vibration control and maintenance aspects.	4
9	Analyze auxiliary drive systems in engines and discuss their role in improving vehicle performance, safety and user comfort.	5
10	Compare gear, chain and belt driven camshaft systems considering reliability, noise, maintenance and cost.	5
	Module III	
11	Draw theoretical and actual valve timing diagrams and analyze their influence on engine performance.	5
12	Compare L, T, F and I head engines considering design features and application in automobiles.	4
13	Prepare comparison chart of SOHC and DOHC systems considering performance, complexity, maintenance and cost.	4
14	Study Variable Valve Timing (VVT) systems and evaluate their role in improving engine efficiency and reducing emissions.	4
15	Prepare a sketch or simple model of poppet valve assembly and present its working through a small group discussion.	4
	Module IV	
16	Compare splash and pressure lubrication systems considering efficiency, reliability and environmental aspects.	4
17	Study classification of engine oils and additives and discuss their role in engine protection and sustainability.	4
18	Sketch lubrication system layout and explain its importance in improving engine life and reliability.	5
19	Study oil pump types (gear, vane, rotor) and compare their performance and maintenance requirements.	4
20	Compare air cooling and water cooling systems considering efficiency, environmental impact and maintenance requirements.	5
	Total Self-Learning hours	90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Automobile engineering vol 2	Kripal singh	Standard
2	Basic Automobile Engineering	C.P. Nakra	Dhanpat Rai Publishing
3	A Textbook of Automobile Engineering	R.B. Gupta	Satya Prakashan

REFERENCE			
Sl. No.	Title	Author	Publication
1	Automotive Technology: A Systems Approach	Jack Erjavec & Rob Thompson	Pearson Education
2	Automotive Mechanics	William H. Crouse & Donald L. Anglin	McGraw Hill Education
3	Internal Combustion Engines	V. Ganesan	Tata McGraw Hill

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4	Automobile Engineering	P.S. Gill	S.K. Kataria & Sons

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://nptel.ac.in/courses/112/104/112104033	All
2	https://www.youtube.com/watch?v=9KKflYch1FE	I & II
3	https://www.engineeringexplained.com	III & IV
4	https://www.howacarworks.com	I & II

Assessment Methodology – Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

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Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60